

TABLE II
YEAR-WISE ROADMAP TO ACHIEVE THE TARGET OF 100GW SOLAR CAPACITIES BY 2020

	2015-16	2016-17	2017-18	2018-19	2019-20	Total
Rooftop solar (GW)	3	7	9	10	11	40
Large-scale solar (GW)	4	6	8	10	12	40
Ultra-mega power projects (GW)	0	5	5	5	5	20
Total (GW)	7	18	22	25	28	100

(Source: RE-Invest by Ministry of New and Renewable Energy)

try and users. A centralised system to disburse capital subsidies may further streamline the process. Introduction of an audit system for monitoring the obligatory solar purchase by discoms and heavy penalty for defaulters may ensure adherence to fulfillment of solar purchase obligation.

It is hardly possible to achieve the ambitious target of 100GW by floating tenders. Therefore just like wind energy segment, feed-in tariff mechanism could be useful in solar as well. The decision to drop anti-dumping duty on the import of solar cells may be a big hindrance for domestic manufacturing.

Growth of solar industry in India opens a huge market for polysilicon. In the absence of domestic capacity, India is heavily dependent on imports. Polysilicon manufacturing involves high initial investment and also needs large volumes to achieve return on investment.

Local manufacturing needs special incentives to compete with global manufacturers. These incentives may either be in the form of tax sops or policies that stipulate special preference for local vendors. This, in turn, will benefit the solar ecosystem in terms of productivity, foreign trade, employability and so on. One good news is that, Adani and SunEdison have announced an investment of US\$ 4 billion in Gujarat for vertically-integrated polysilicon refining facility.

Need to facilitate investment

Cost and financing are the key hurdles in the way of realising the 100GW target. Potential problems

with transmission and land acquisition could also act as hurdles in the way of achieving the mammoth targets.

Specific solar sector schemes need to be implemented for reducing the cost of finance and increasing the return on investment. Sectoral cap for lending in the power sector is also applicable for solar and other renewable energy sectors. Raising finance for solar systems is a challenge with a dearth of low-interest loans due to this sectoral cap. Removal of this cap for renewable energy sector will improve leverage for the renewable industry.

Benchmark prices offered under different government subsidy schemes or tenders for solar system installation are often considerably lower than actual costs of project implementation. This leads to disinterest among industry players about government tenders. Public-private partnerships for government-owned solar projects involving local entrepreneurs as well as electricity supply companies can be an effective model to ensure industry participation. Various forms of incentives such as tax holidays and partial risk guarantee (from state governments)

can be provided to encourage the solar industry.

Moreover, given that solar plants tract huge demand for land may turn out to be a challenge unless the draconian land acquisition laws in the country are streamlined. Effective planning of projects also requires site-specific pre-feasibility analysis and assessment of local household and agricultural demand in the area.

Efficient storage in demand

Availability and reliability are imperatives for any successful energy source. Energy-storage systems and charging mechanisms play a critical role to ensure reliability and efficiency of solar power sources in the country, especially during the season of low solar light and in the hinterlands of the country that are prone to frequent power blackouts.

An efficient energy storage system also helps ensure reduction of maintenance cost by countering intermittency in off-grid systems. According to an industry source, more than 50 per cent of component cost for any off-grid solar installation accounts for energy storage systems. Technological innovations are required to develop cheaper and more efficient energy storage systems.

In the last decade, lithium-ion based energy storage systems have helped promote implementation of off-grid solar energy facilities by ensuring a captive source for continuous power supply, thereby minimising dependency on fossil fuel or on-grid systems.

India's solar rooftop map for 2016

1. Over 525MW of rooftop solar projects have been commissioned as of October 31, 2015
2. Tamil Nadu is the leading state with rooftop installations of more than 76MW
3. Total installed capacity in the residential segment is 143MW or 27 per cent of the total market
4. Residential market is highly fragmented and dominated by local companies
5. Industrial and commercial markets will lead the growth of rooftop solar installations going forward

(Source: BRIDGE TO INDIA)